HACKCULT '26 / PS2

HeritageLens

AI-powered cultural archive for India's architectural memory

A working system for digitization, OCR, search, maps and source-grounded research

WORKING PROTOTYPE

TEAM HERITAGELENS // IIT GANDHINAGAR



The problem and what the system must achieve

Heritage evidence exists, but its formats and archives cannot work together.





Knowledge stays fragmented

Photographs lack metadata. Scanned documents hide their text. Field notes and regional records remain disconnected from the monuments they describe.

CORE REQUIREMENTS

01 **Preserve originals**

Durable private storage and authorized downloads

02 **Make text searchable**

PDF extraction and OCR for scanned sources

03 **Keep human oversight**

Curator review before approval

04 **Connect context**

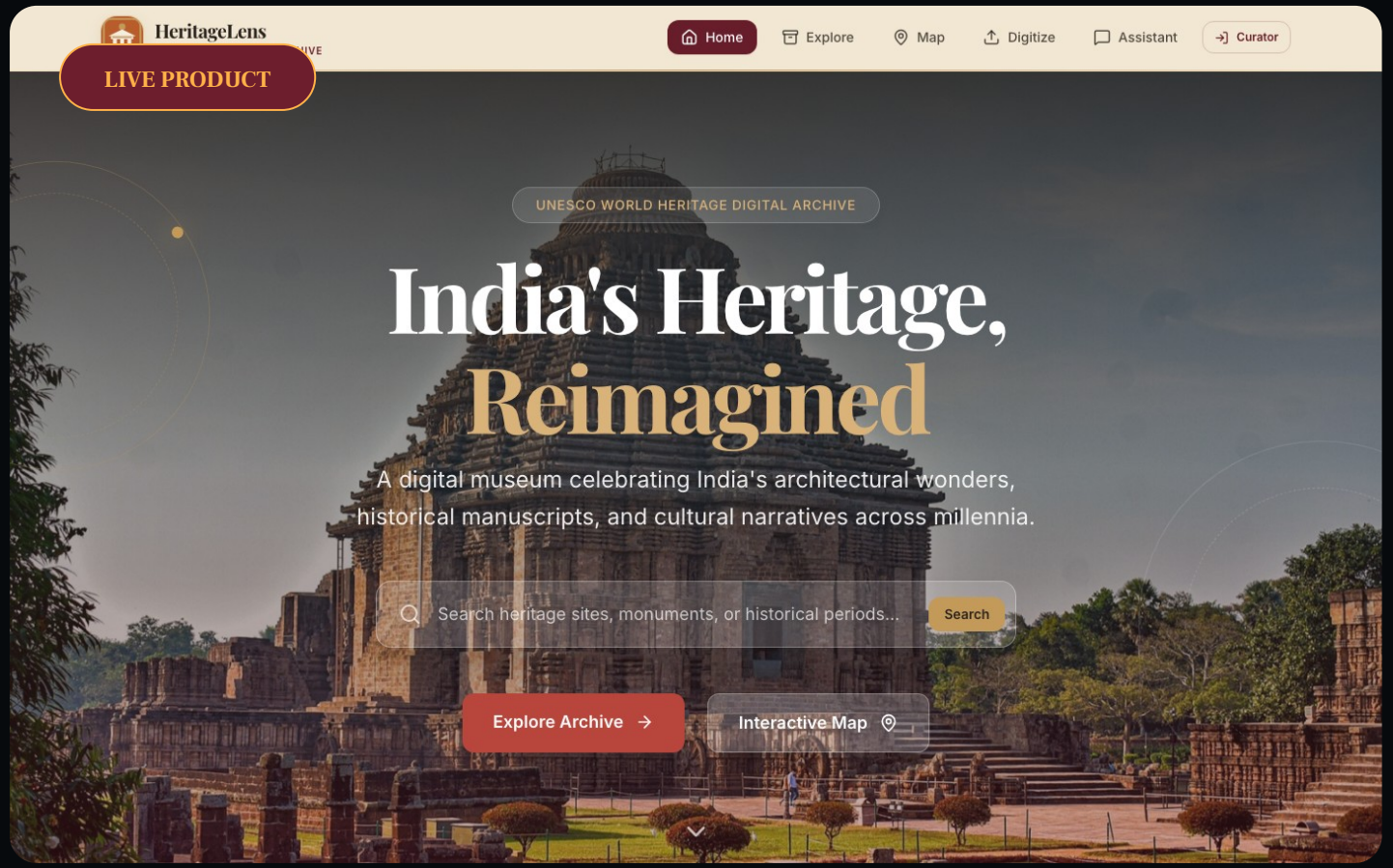
Sites, documents, images and map coordinates

05 **Support public discovery**

Open browsing, filtering and grounded answers

HeritageLens turns scattered sources into a usable archive

The judges can inspect the deployed product, not a mock-up.



01 **Searchable archive**

Site metadata plus extracted and corrected document text

02 **Document digitization**

PDF extraction, Tesseract OCR and curator correction

03 **Interactive map**

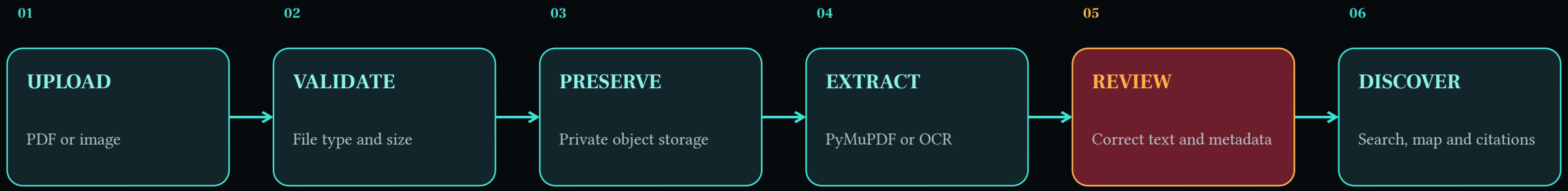
Coordinates connect monuments to archival records

04 **Grounded assistant**

Answers reference retrieved archive sources

From an uploaded source to verified discovery

Automation handles repetitive processing. Curators retain editorial control.



PERSISTED PROCESSING STATES

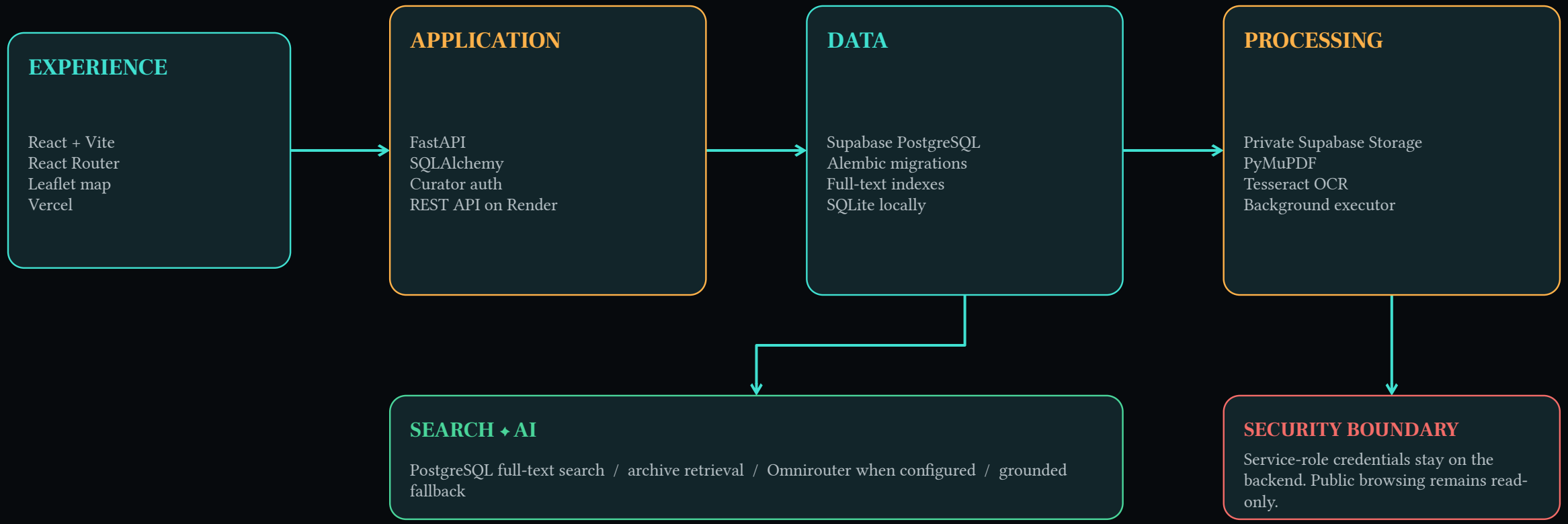
UPLOADED · PROCESSING · EXTRACTED · NEEDS REVIEW · APPROVED · FAILED

Failure remains visible and recoverable

The backend records errors, preserves the original source and can resubmit interrupted background work after restart.

Technical architecture and stack

A modular monolith keeps development fast without mixing responsibilities.



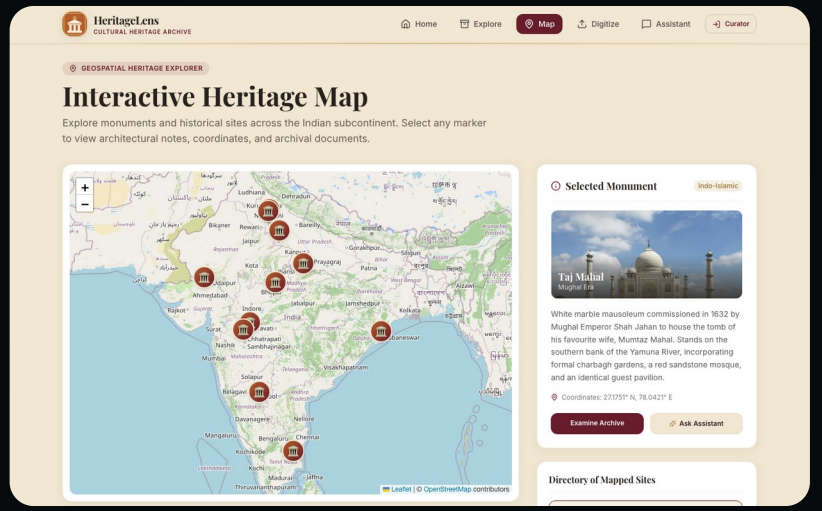
Feasibility is demonstrated by the working build

The architecture uses services already configured for the deployment target.

11
seeded heritage records

39
backend tests passed

3
browser workflows passed



WHY IT CAN SHIP

1. Vercel frontend and Render backend already deployed
2. Supabase Session Pooler and private object storage supported
3. Alembic controls schema changes without startup mutations
4. CI checks migrations, backend tests, lint, build and browser flows

Known limits

Shared curator credential for the prototype. In-process OCR uses one Render worker. Distributed queues and individual roles remain future work.

Impact now, stronger retrieval next

The current product solves preservation and discovery first. Phase 2 can improve multilingual retrieval after production verification.

Researchers

Find evidence across sites and digitized records

Archivists

Preserve originals and approve corrected text

Institutions

Publish structured collections with traceable sources

Visitors

Explore monuments through maps and accessible narratives

NEXT

01

Production verification

Apply migrations, configure private storage and test live upload

02

Multilingual retrieval

Add embeddings and hybrid ranking for regional-language archives

03

Evaluation

Measure retrieval quality and citation faithfulness

HeritageLens makes cultural evidence searchable without losing the path back to its source.